

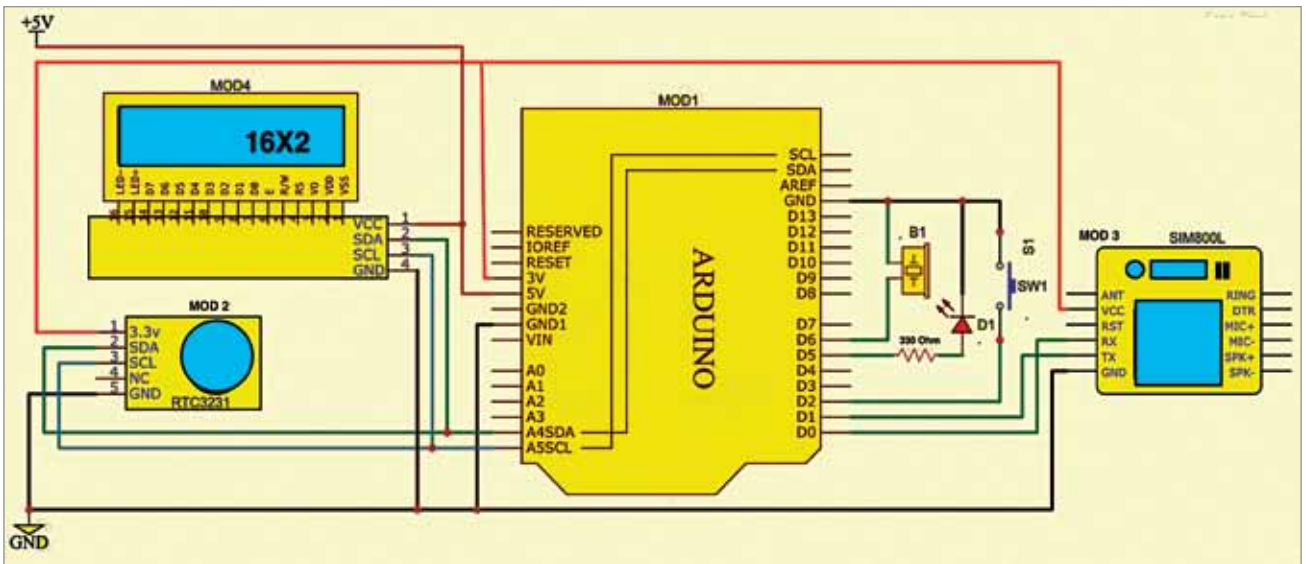


Fig. 2: Circuit diagram

```

1 #include <Wire.h>
2 #include <RTClib.h>
3 // #include <LiquidCrystal.h>  /// Uncomment for SPI 8 Wire LCD
4 // LiquidCrystal lcd(7, 8, 9, 10, 11, 12);  /// Uncomment for SPI 8 Wire LCD
5 #include <LiquidCrystal_I2C.h>  // For I2C based display
6 LiquidCrystal_I2C lcd(0x27,20,4);  // set the LCD address to 0x27 for a 16 chars and 2 line display
7 // Initialize RTC and LCD
8 RTC_DS3231 rtc;
9 // Define pins
10 #define BUZZER_PIN 6
11 #define BUTTON_PIN 2
12 #define LED_PIN 5
13 // Buzzer control variables
14 bool buzzerState = false;  // Tracks whether the buzzer is ON
15 unsigned long buzzerTimer = 0;  // Tracks the time the buzzer started
16 bool buttonPressed = false;  // Tracks if the button was pressed
17 bool alarmTriggered = false;  // Tracks if the current alarm has been handled
18 bool buzzerReactivating = false;  // Tracks whether the buzzer is in the reactivation cycle
19 // Number of reactivations before sending SMS
20 int reactivationCount = 0;
21 // Array for days of the week
22 const char* daysOfTheWeek[] = {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
23 // Schedule times and messages
24 struct Alarm {
25   int hour;
26   int minute;
27   const char* message;

```

Fig. 3: Snippet of the source code

cable from the board and assemble the components as per the circuit diagram (see Fig. 2).

Now, connect a 12V, 2A adaptor to the Arduino Uno DC socket. The gadget then operates as an electronic calendar, displaying the date, time, day, and year.

Whenever a scheduled task time arrives, the LCD displays the corresponding message and the

buzzer sounds.

If the user presses the button, the device resumes standard calendar functionality. However, if the button is not pressed after two reactivations, an SMS notification is sent to the predefined phone number in the code.

Note. This device demonstrates how it can serve as a medicine reminder for patients and a time-table reminder for students. Users can customise the schedule and messages

displayed on the LCD as per their preferences. **EFY**

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 to access and download source code
 and other resources for this project

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